



Create an Android-friendly Hotspot in Linux

This article walks readers through the process of setting up an Android-friendly hotspot.

Let's suppose you have a laptop and a phone, and you want to share your laptop's Internet connection with the phone (reverse tethering). You might think it's as simple as going to a network connection and creating a wireless connection. Unfortunately, it's not, because the hotspot created by laptops generally supports only two modes—Ad-Hoc and Infrastructure (which can be seen in Figure 1). Both these modes are not supported by most of the Android and Windows phones in the market. So let us go about creating an Android-friendly or a 'phone-friendly' hotspot. For this article, I'll be using Ubuntu 12.04 LTS.

Installation

Let's first install Hostapd from the Ubuntu software centre.

If you prefer the command line, then just type the following command:

```
sudo apt-get install hostapd
```

You can also download and compile Hostapd from source, from its site <http://w1.fi/hostapd/>.

For other distributions, search for the package named *hostapd* in your repository.

As explained in <http://acx100.erley.org/git.html>, Hostapd has the following levels in managing Wi-Fi and each level is explained as follows:

“**Wireless card:** Should be self-explanatory... handles

sending/receiving wireless packets

Linux kernel: The Linux kernel contains the driver for the wireless card, and the *nl80211* sub-system, which handles the configuring of wireless interfaces for user space

libnl: *libnl* is the transport layer used to communicate with the kernel via netlink

udev: *udev* is the facility the kernel uses to pass events/calls to *crda*

iw: *iw* is a userspace utility that we can use to test that *libnl* is working correctly, as well as to create additional virtual wireless interfaces on the wireless card

crda: *crda* is a userspace program that the kernel queries (through *udev*) to find what channels/frequencies are usable, and at what powers. This moves the information out of static tables maintained in the kernel to userspace, which allows them to be updated without reloading drivers or rebooting

Wireless regulatory database: This is the database of allowable frequencies and transmit power levels used by *crda*

Hostapd: This is the daemon that handles the generation of beacons and other wireless packets, as well as *wpa-psk*, *wpa2*, etc, encryptions.”

Checking your Wi-Fi card

Hostapd supports the following drivers:

- *mac80211*-based drivers with support for master mode [*linux*]
- Linux drivers that support *nl80211/cfg80211* in AP mode